Beyond Ingress

Mastering the Kubernetes Gateway API

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Agenda

Expose pods

Gateway API

Routing

Gateway API vs. Ingress

Gateway Controller

Demo

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Expose Pods

Expose Pods

Service with type Load Balancer

Ingress

Gateway API

Ingress and Gateway API Features

- One entry point
- Better routing
- Automate TLS certificates with Cert-Manager
- TLS termination

Ingress

Ingress Controller:

- Traefik
- Nginx
- Application Gateway Ingress Controller (AGIC)

Ingress Downsides

Ingress controllers have a lot of control how traffic is routed

Annotations are unique to the used Ingress controller (e.g. Nginx) → vendor lock-in

No cross-namespace support for TLS secrets

Ingress Downsides

Ingress does not support traffic policies

No modularity

No real Separation of Concerns

Gateway API GA in 2023 to solve these downside

Gateway API

Gateway API

Components are modular and split into multiple parts

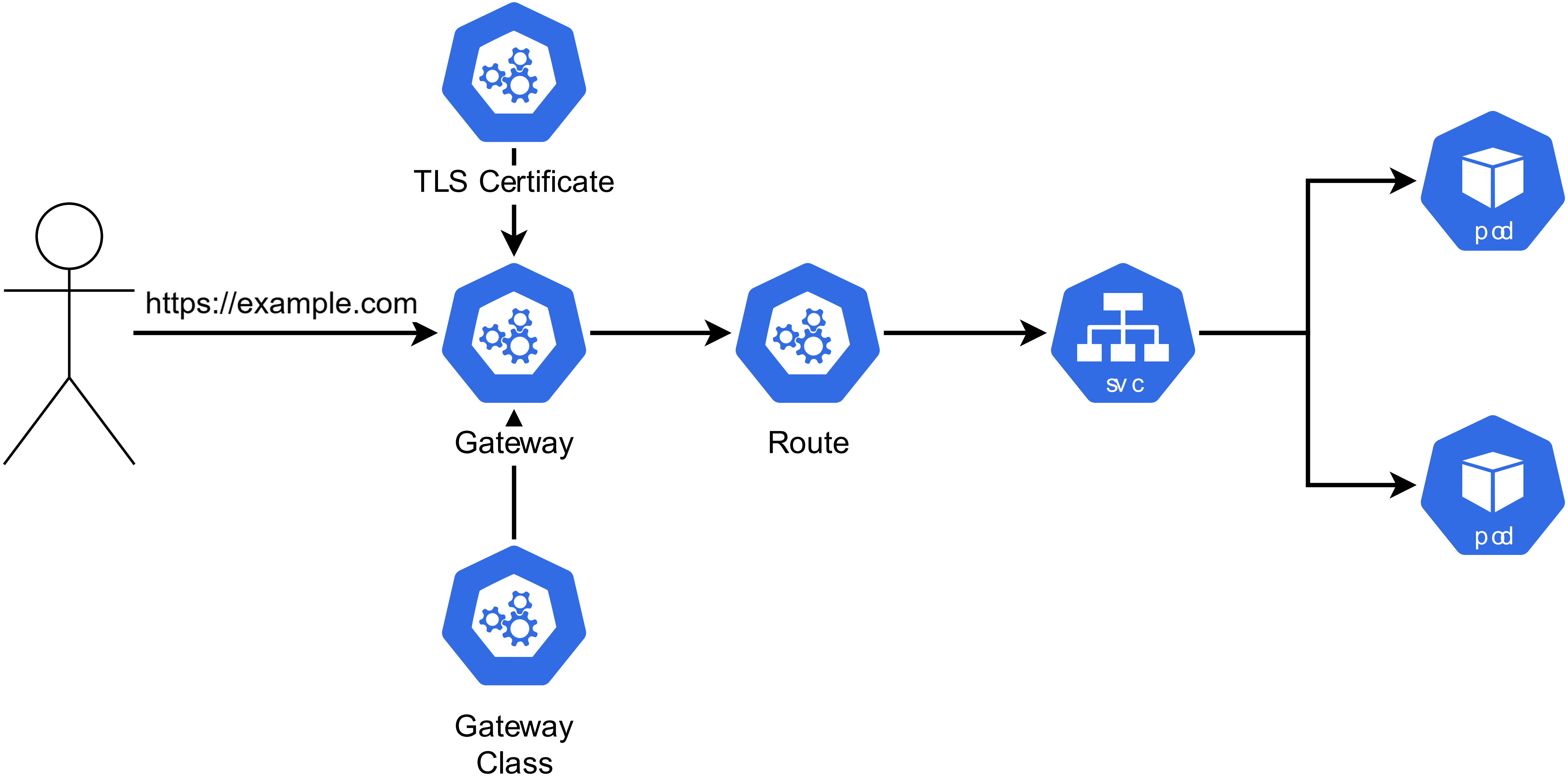
Gateway Class configures proxy (similar to Ingress Class)

Gateway Class depends on Gateway Controller

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass
metadata:
  name: envoy
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass
metadata:
  name: envoy
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass
metadata:
  name: envoy
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```


TLS Certificate
https://example.com
Gateway
Route
svc
pod
pod
Gateway
Class

Gateway

Gateway contains listeners

Governs TLS certificates

Namespaced

Gateway consumes global Gateway Class

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: Gateway
metadata:
  name: Gateway
  namespace: gateway-ns
  annotations:
    cert-manager.io/cluster-issuer: clusterissuer
spec:
  gatewayClassName: envoy
  listeners:
  - name: http-listener
    port: 80
    protocol: HTTP
    allowedRoutes:
      namespaces:
        from: All
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: Gateway
metadata:
  name: Gateway
  namespace: gateway-ns
  annotations:
    cert-manager.io/cluster-issuer: clusterissuer
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spec:
  gatewayClassName: envoy
  listeners:
  - name: http-listener
    port: 80
    protocol: HTTP
    allowedRoutes:
      namespaces:
        from: All
```

```yaml
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```

```yaml
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```

```yaml
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```

Routing

Gateway Routing

Listeners can listen on different ports

Different protocols:

- HTTP
- HTTPS
- TLS
- TCP
- UDP

Routes

Weighted load balancing (traffic splitting)

Uses standards instead of annotations

Routes:

- HTTPS and HTTP
- TCP and UDP
- GRPC
- TLS

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
      backendRefs:
        - name: traefik
          port: 80
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
      backendRefs:
        - name: traefik
          port: 80
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
      backendRefs:
        - name: traefik
          port: 80
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
      backendRefs:
        - name: traefik
          port: 80
```

Routing

Routing options:

- Path
- Header, e.g. User-Agent
- Query
- HTTP method
- Redirect request

```yaml
hostnames:
  - "demo.programmingwithwolfgang.com"
rules:
- backendRefs:
  - name: app-one
    port: 80
    weight: 20
  - name: app-two
    port: 80
    weight: 80
```

```yaml
hostnames:
  - "demo.programmingwithwolfgang.com"
rules:
- backendRefs:
  - name: app-one
    port: 80
    weight: 20
  - name: app-two
    port: 80
    weight: 80
```

```yaml
rules:
  - matches:
      - path:
          type: PathPrefix
          value: /my/path
    filters:
      - type: URLRewrite
        urlRewrite:
          path:
            type: ReplaceFullPath
            replaceFullPath: /replaced
    backendRefs:
      - name: app-one
        port: 80
  - backendRefs:
    - name: app-two
      port: 80
```

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rules:
  - matches:
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    - name: app-two
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  - matches:
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      - name: app-one
        port: 80
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    - name: app-two
      port: 80
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rules:
  - matches:
      - path:
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      - type: URLRewrite
        urlRewrite:
          path:
            type: ReplaceFullPath
            replaceFullPath: /replaced
    backendRefs:
      - name: app-one
        port: 80
  - backendRefs:
    - name: app-two
      port: 80
```

Gateway API vs Ingress

Gateway API vs Ingress

Gateway API might not replace Ingress

Ingress is very easy to manage

- Especially in smaller teams

Gateway might be better for multi-team setups due to modularity

Gateway Controller

Gateway Controller

Nginx Gateway Fabric

Traefik

Envoy Gateway

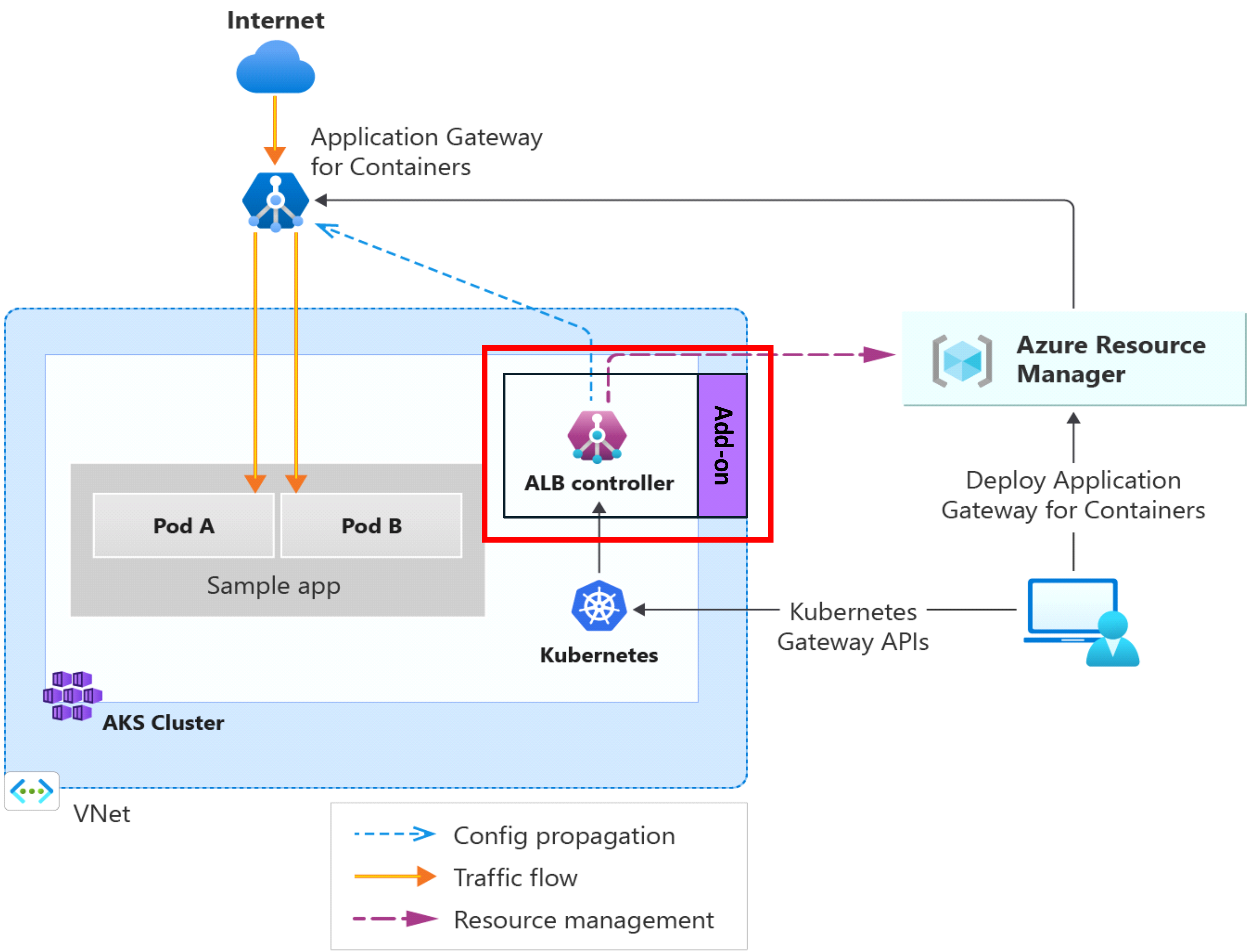
Azure Application Gateway for Containers

Application Gateway for Containers

AKS addon-on in preview

Brings additional features:

- Advanced health checks
- WAF support
- Monitoring
- Autoscaling and high-availability


Internet
Application Gateway
for Containers
Azure Resource
Manager
Add-on
ALB controller
Pod A
Pod B
Sample app
Deploy Application
Gateway for Containers
Kubernetes
Gateway APIs
Kubernetes
AKS Cluster
VNet
Config propagation
Traffic flow
Resource management

Demo

Thank you for watching!

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